

Pre-Class Response for Lecture 11:

Suppose we are doing a randomized control trial for a new drug, Stanfordizone. (It treats the compulsive urge to found a tech start-up; side effects may include wearing hoodies to formal events). We get a representative sample of 1000 people to participate, and split the sample randomly into a control group of 500 people and a treatment group of 500 people.

Unfortunately, after the experiment is over, follow-up surveys reveal that only about 400 people in the treatment group actually took the drug. (The other 100 people were presumably too busy with their start-ups and forgot).

The research team brainstorms and comes up with the following options:

1. The study was compromised, and we can't learn anything about the effect of Stanfordizone. We should throw out the data and re-do the study with better monitoring.
2. We should throw out the data from the 100 people who didn't take the drug, and analyze the rest of the data as normal. (That is, do "average among treatment group - average among control group" among the remaining 900 people). That should give us that ATE of Stanfordizone on start-up-starting, but the error bars might be larger since we have a slightly smaller sample.
3. We can't learn the ATE of Stanfordizone on start-up-starting on the original population we sampled the 1000 study participants from, but we may still be able to learn something from our data. We should think further about this.

What do you think about each of these options? For option 3, what do you think we can learn from this study, if anything? Can we get close to the ATE on the original study population? Or is there some other population for whom we can estimate the average treatment effect of Stanfordizone?